

18 May 2026

Dear Nature Biotechnology Editors,

Subject: Submission of "Foundation-model morphology maps residual programs in spatial-omics contours" as a Brief Communication

Please consider this manuscript for publication as a Brief Communication in Nature Biotechnology.

This submission is prepared for double-anonymous peer review. The reviewer-facing manuscript, Online Methods and Supplementary Information omit author identity and affiliation; code-repository links are included in the manuscript Code availability section so editors and reviewers can access the software, fixed code release and released upload package.

The timing of the study is important. Atera WTA is a newly introduced whole-transcriptome in situ platform for FFPE tissue that moves spatial profiling from targeted panels toward more than 18,000 genes in tissue context. The public breast and cervical cancer preview datasets used here profile 18,028 genes and, to our knowledge at submission, are the publicly available Atera WTA preview examples for epithelial cancer. This creates a timely analysis problem for the journal's readership: how should co-registered H&E foundation-model morphology be interrogated when spatial omics already defines transcriptome-scale molecular contours?

The manuscript introduces morphomolecular translation mapping (mTM), a contour-constrained framework for asking whether H&E foundation-model embeddings map residual WTA program variation after spatial-omics-derived contour labels and spatial covariates have already been fixed. This is not an H&E-to-expression leaderboard. The primary question is whether morphology carries measurable molecular information within already defined WTA contours.

In breast Atera WTA, mTM nominates residual luminal estrogen-response, unfolded-protein-response and oxidative-phosphorylation continua inside S3 contours. In cervical Atera WTA, the same residual-decoding workflow serves as a second epithelial cancer stress test and nominates stromal-remodeling, immune-ecology and invasion-associated program families rather than a direct replication of the breast luminal phenotype. The reported associations are tested against compartment-aware spatial permutations that preserve contour labels, centroid-position bins and boundary-distance bins; 13 candidate associations pass the 95% spatial-null gate.

The study is deliberately conservative about what the data support. The contours were generated from Atera WTA cell-coordinate and cluster information rather than from H&E segmentation, so the manuscript does not claim independent histological contour discovery, morphology-only deployment, causality, directional boundary mechanisms or protein-level validation. We believe the Brief Communication format is appropriate because the manuscript makes one focused point: foundation-model morphology can add residual information within spatial-omics-defined contours rather than merely reproduce those contours.

Editor-only information:

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Code availability for editors: code identifiers are provided here and in the manuscript Code availability section to support editorial and reviewer assessment.

pyXenium repository: <https://github.com/hutaobo/pyXenium>

Fixed pyXenium release: <https://github.com/hutaobo/pyXenium/releases/tag/v0.4.6>

Submission-packaging hardening commit:

<https://github.com/hutaobo/pyXenium/commit/3c4617010bb1c76e18921958c4507f91512aa9dc>

Journal-upload package and individual submission files:

<https://github.com/hutaobo/pyXenium/releases/tag/nbt-submission-upload-20260518-draft>

HistoSeg contour-generation software: <https://github.com/hutaobo/HistoSeg>

Author identity and affiliation are provided in this cover letter for editorial use.

Submission status: this is an initial submission to Nature Biotechnology. The manuscript has not been published elsewhere and is not under consideration by another journal.

Competing interests and author responsibility: the author declares no competing interests, has approved the manuscript and agrees with its submission. Taobo Hu conceived the study, implemented the analysis workflow, performed the computational analyses, generated the figures and source data, and wrote the manuscript.

Ethics and data use: the study used public/vendor example spatial WTA and H&E data and did not involve new human sample collection, intervention or generation of new identifiable participant data.

Sincerely,

Taobo Hu